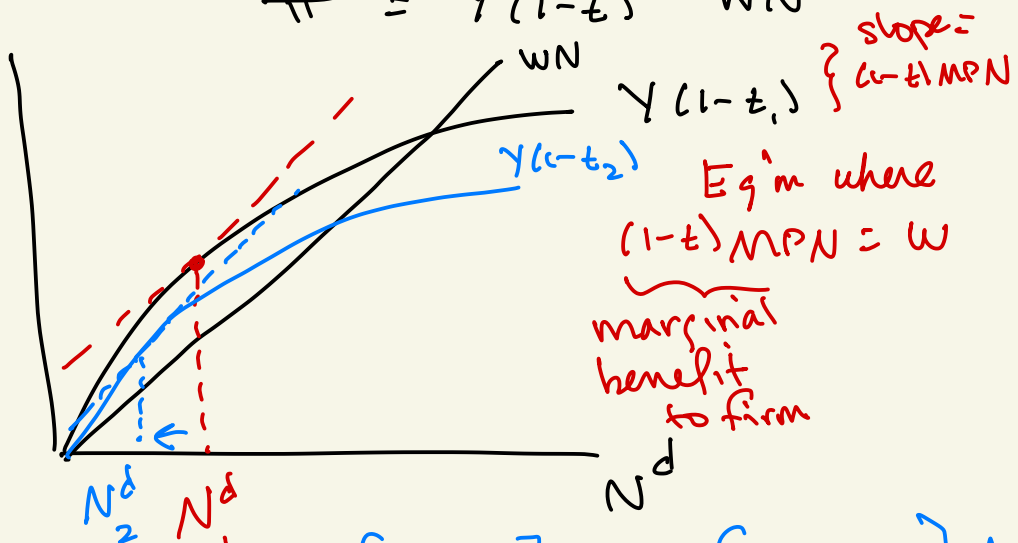



Taxation : Representative Firm (N^d)

Ex #1, Production tax

$$\pi = Y(1-t) - wN^d$$



Assume $t \uparrow \Rightarrow [Y(1-t)] \downarrow \Rightarrow [(1-t)MPN] \downarrow$

$$(1-t)MPN = \bar{w}$$

no Δ

by policy $\downarrow$ need $MPN \uparrow$ in equilibrium } choose $N^d \downarrow$ to set $MPN \uparrow$

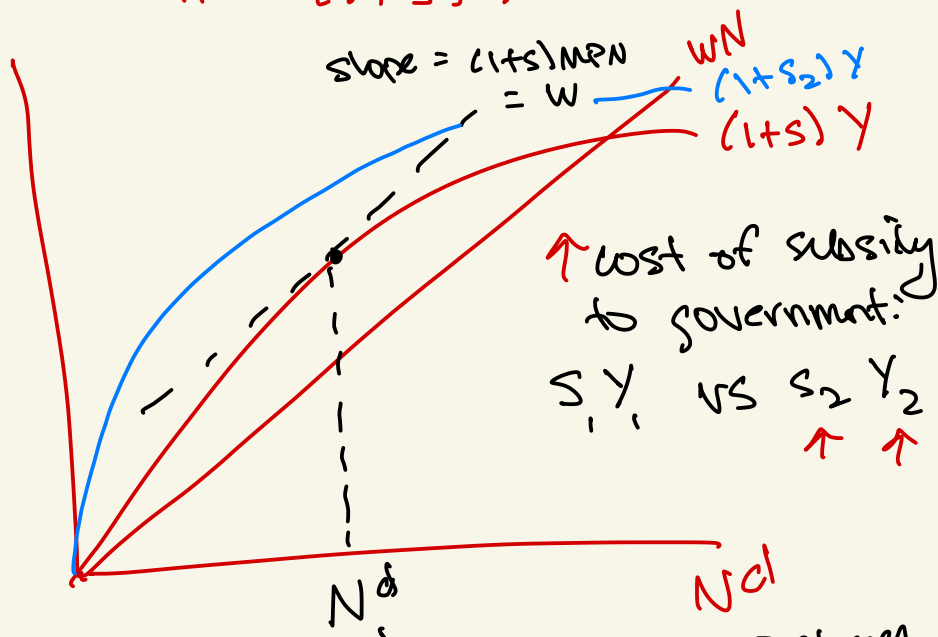
flatter slope

$$N \downarrow \quad Y \downarrow$$

Tax Revenue : $t \cdot Y$ } Δ ?
 $\uparrow \quad \downarrow$ } ambiguous

Ex #2 Production subsidy (negative tax)

$$\pi = (1+s)Y - wN$$



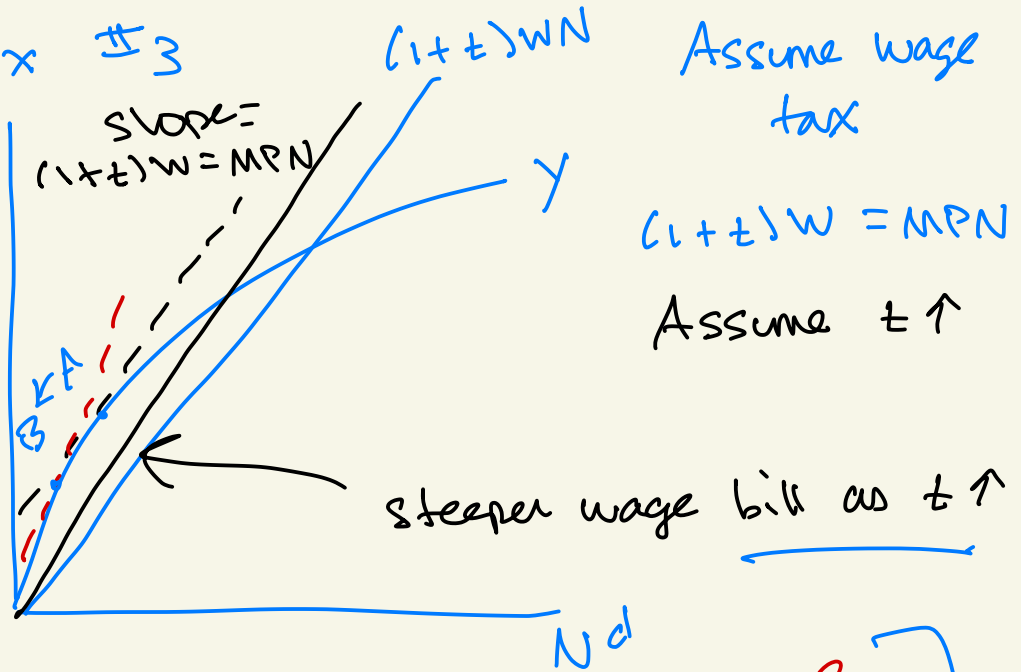
Assume $s \uparrow \Rightarrow \underbrace{(1+s)MPN}_{\uparrow}$ } steeper

$$\underbrace{(1+s)}_{\uparrow} \underbrace{MPN}_{\downarrow} = \underbrace{w}_{\text{no } \Delta}$$

in equilibrium $\Rightarrow$ firms choose $N \uparrow$

$$\therefore Y \uparrow$$

Ex #3



↑ tangency at steeper part of production function

$$MPN = (1+t)W$$

need ↑ no Δ

$MPN \uparrow$ so
lay off some unproductive
workers $N \downarrow \rightarrow Y \downarrow$

$$\Delta Rev : t \bar{W} N \quad \left. \begin{array}{c} \uparrow \quad \downarrow \end{array} \right\} ?$$

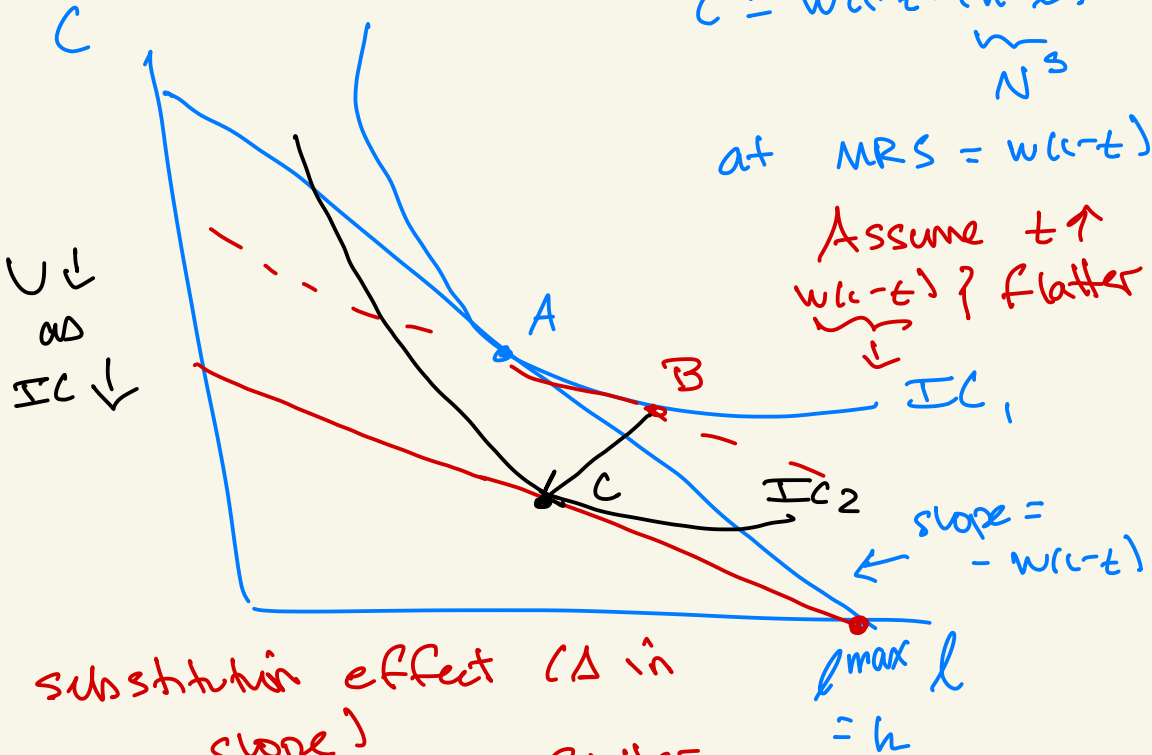
Taxation : Rep household

Assume $\pi - \tau = \phi$

$$C = \underbrace{w(l-t)}_{N^S} (h-l)$$

at $MRS = w(l-t)$

Assume $t \uparrow$
 $w(l-t) \uparrow$ flatter
 $\downarrow$



substitution effect (Δ in slope)

$A \rightarrow B$
 $C \downarrow, l \uparrow$
 $\left\{ \begin{array}{l} MRS \downarrow \\ \uparrow \frac{MU(l)}{MU(C)} \end{array} \right.$
 $= w(l-t_2)$ flatter

Income effect (lower than before)
 $C \downarrow, l \downarrow$
 $\underbrace{\quad}_{\text{normal goods}}$
 $\Rightarrow N \uparrow$ if $l \downarrow$

$\Delta C < 0$ from sub and income effects
 $\Delta l ? \therefore \Delta N^S ?$
 $\Delta Rev : t \cdot \bar{w} \cdot N^S$
 $? \quad \uparrow \quad ?$

Using math to forecast ΔN^S

Assume log utility

$$U(c, l) = a \ln c + b \ln l$$

$$C = w(1-t)(h-l)$$

$$\text{Set } \pi - \tau = \phi$$

To find equilibrium

sub effect $\rightarrow$ ① Set $MRS = w(1-t)$

$$\frac{b/l}{a/c} = \frac{MU(l)}{MU(c)}$$

$$\frac{b}{a} \frac{c}{l} = w(1-t)$$

both sub and income effects

② Plug in the budget constraint

$$\frac{b}{a} \frac{w(1-t)(h-l)}{l} = w(1-t)$$

income effect $\uparrow$

$$\frac{b}{a} (h-l) = l$$

$$\frac{b}{a} h - \frac{b}{a} l = l$$

$$\frac{b}{a} h = \left(\frac{a}{a} + \frac{b}{a}\right) l$$

$$l^* = \frac{b}{a+b} h$$

$$N^* = h - \frac{b}{a+b} h = \frac{a}{a+b} h$$

don't depend on $w(1-t)$!!

with log utility :

$$\textcircled{1} \Delta \text{Rev} = \Delta [t \cdot \underbrace{w \cdot N^S}_{\text{no } \Delta}]$$

$\uparrow \qquad \qquad \qquad \uparrow$

$\textcircled{2}$ Income and Substitution effects exactly cancel for l and N^S

What about C ?

$$C = w(1-t)(h-l) = w(1-t)N^S$$

$$C = w(1-t) \frac{a}{a+b} h$$

$\underbrace{\qquad\qquad\qquad}_{\text{as}} \quad w(1-t) \uparrow \quad C \uparrow$
 $w(1-t) \downarrow \quad C \downarrow$

While we predict from income and sub effects

Try quasi-linear utility

$$U = ac + b \ln l$$

$$w(1-t)(h-l) = C$$

(1) Sub effect: $MRS = w(1-t)$

$$\frac{b/l}{a} = \frac{MU(l)}{MU(c)} = w(1-t)$$

$$\frac{b}{a} \cdot \frac{1}{l} = w(1-t)$$

can't plug in for income

$$\left\{ \frac{b}{a} \cdot \frac{1}{w(1-t)} = l^* \leq h \right.$$

(pure substitution effect)

as $w(1-t) \uparrow$ $l \downarrow$

$N \uparrow$
upward-sloping N^S

in data over business cycles
 $w(1-t)$ and N are positively
correlated (Ch 3)

$$C = w(1-t)(h-l)$$

$$= w(1-t)h - \cancel{w(1-t)l} - \frac{b}{a - \cancel{w(1-t)l}}$$

$$C = w(1-t)h - \frac{b}{a}$$

$$\text{as } [w(1-t)] \uparrow \quad C \uparrow$$

from both income and
substitution effects
(as in the graphs)

We choose
quasi-linear utility $\Rightarrow$ to
explain short term
(business cycle) movements
in L and N in
response to $\Delta [w(1-t)]$

log utility $\Rightarrow$ to explain why
long-term labor force
participation and hours
worked don't move much
even $w(1-t)$ does.

* Preferences seem different
over long run than short
run